SWAYAM NPTEL COURSE ON

STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 11: Random Graph

Lecture 01: Network Data and Random Graph

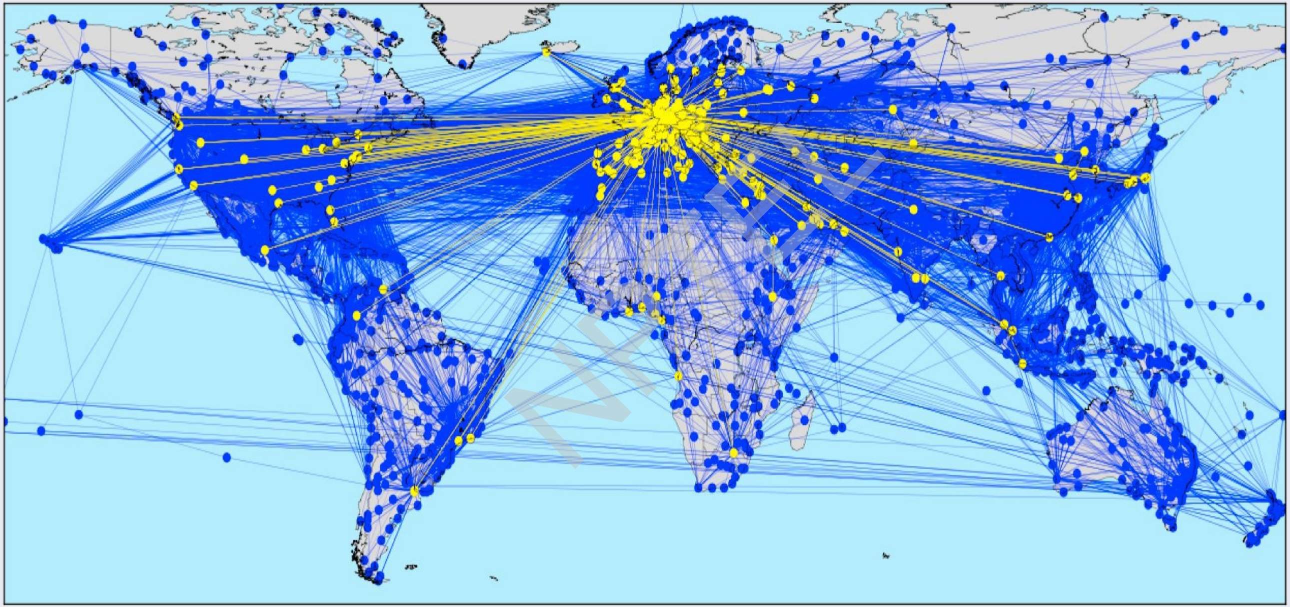
Examples of Network Data

- **Social Networks**
  - Facebook friendship network
  - Twitter follower network
  - Instagram interaction network
- **Biological Networks**
  - Protein–protein interaction networks
  - Gene regulatory networks
  - Neural connectivity
- **Communication Networks**
  - Email exchange networks
  - Phone call networks
  - Internet routing topology

Examples of Network Data (continued)

- **Technological Networks**
  - Power grid network
  - Transportation networks (airline routes, road networks)
  - Sensor networks
- **Information Networks**
  - Web hyperlink network
  - Citation networks
  - Recommendation networks
- **Economic Networks**
  - Trade networks between countries
  - Supply-chain and logistics networks
  - Interbank lending networks

Example of Network Data (continued)



(Source: https://doi.org/10.1016/j.ssci.2017.11.010)

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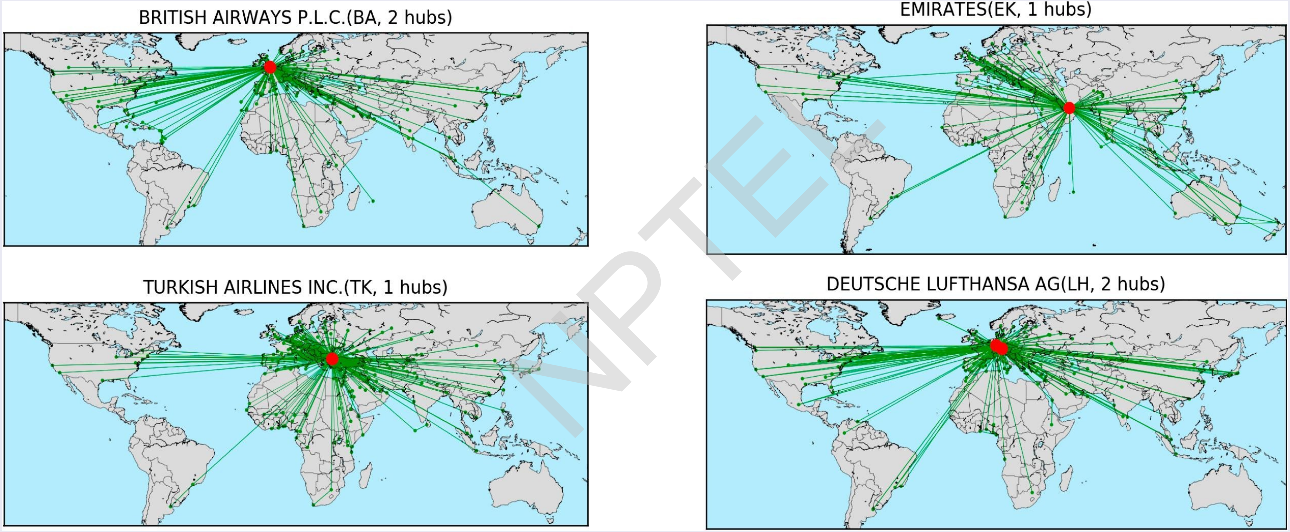

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Picture of Network Data (continued)

ID	IATA	Name	Passengers	Nodes	Links	Density	ASPL	ADEG
1	AA	AMERICAN AIRLINES INC.	13.19 Mio (4.03%)	186	1049	3.05%	2.28	11.3
2	DL	DELTA AIR LINES INC.	13.05 Mio (8.01%)	249	1162	1.88%	2.28	9.3
3	WN	SOUTHWEST AIRLINES CO.	13.03 Mio (11.99%)	95	1281	14.34%	1.97	27.0
4	FR	RYANAIR Ltd.	8.78 Mio (14.67%)	180	2438	7.57%	2.20	27.1
5	UA	UNITED AIRLINES INC.	8.66 Mio (17.31%)	193	967	2.61%	2.57	10.0
6	CZ	CHINA SOUTHERN AIRLINES	7.6 Mio (19.63%)	134	994	5.58%	2.39	14.8
7	U2	EASYJET AIRLINE COMPANY L..	6.46 Mio (21.61%)	133	1430	8.15%	2.12	21.5
8	MU	CHINA EASTERN AIRLINES	6.02 Mio (23.44%)	131	853	5.01%	2.34	13.0
9	TK	TURKISH AIRLINES INC.	5.26 Mio (25.05%)	255	648	1.00%	2.09	5.1
10	LH	DEUTSCHE LUFTHANSA AG	5.1 Mio (26.60%)	169	494	1.74%	2.15	5.8
11	EK	EMIRATES	4.87 Mio (28.09%)	130	276	1.65%	2.07	4.2
12	CA	AIR CHINA LIMITED	4.71 Mio (29.53%)	124	538	3.53%	2.21	8.7
13	NH	ALL NIPPON AIRWAYS CO. LT..	4.43 Mio (30.88%)	76	260	4.56%	2.42	6.8
14	AF	AIR FRANCE	3.95 Mio (32.09%)	151	369	1.63%	2.21	4.9
15	BA	BRITISH AIRWAYS P.L.C.	3.83 Mio (33.26%)	183	405	1.22%	2.60	4.4

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Basic Definition of a Graph

A *graph* G is an ordered pair

$$G = (V, E),$$

where:

- V is the set of *vertices*.
- $E \subseteq \{\{u, v\} : u, v \in V, u \neq v\}$ is the set of *edges*.

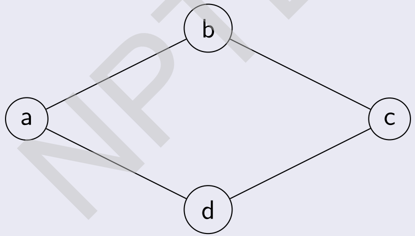
Example:

$$V = \{1, 2, 3\}, \qquad E = \{\{1, 2\}, \{2, 3\}\}.$$

Vertex Set and Edge Set

The **vertex set** $V(G)$ is the collection of nodes. The **edge set** $E(G)$ is the collection of unordered pairs $\{u, v\}$.
Example: For the graph below,

$$V = \{a, b, c, d\}, \qquad E = \{\{a, b\}, \{b, c\}, \{c, d\}, \{a, d\}\}.$$



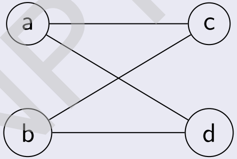
Bipartite Graph

A graph $G = (V, E)$ is *bipartite* if V can be partitioned as

$$V = V_1 \cup V_2, \qquad V_1 \cap V_2 = \emptyset,$$

such that every edge connects a vertex in V_1 to one in V_2.

Example:

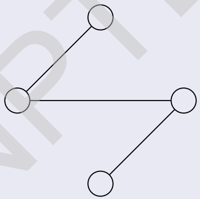


This is a complete bipartite graph $K_{2,2}$.

Tree

A **tree** is a connected graph with no cycles. Equivalently:

- A tree with n vertices has exactly $n - 1$ edges.
- There is a unique path between any two vertices.

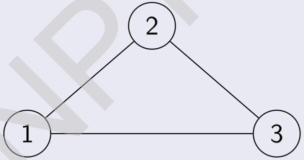


Complete Graph and Clique

A **complete graph** K_n is a graph where every pair of vertices is connected by an edge:

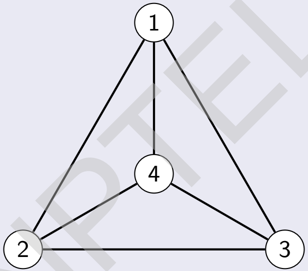
$$|V| = n, \qquad |E| = \binom{n}{2}.$$

A **clique** in a graph is a subset of vertices whose induced subgraph is complete.



This is K_3, a 3-clique.

Clique



This is the example of 4-clique.

Graph Isomorphism

Graphs $G = (V, E)$ and $H = (V', E')$ are **isomorphic** if there exists a bijection

$$\varphi : V \to V'$$

such that

$$\{u, v\} \in E \iff \{\varphi(u), \varphi(v)\} \in E'.$$

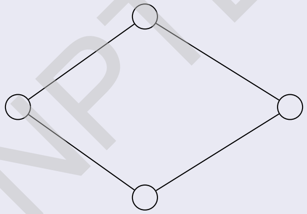
Meaning: the graphs have identical structure, just different labels.

Cycles

A **cycle** C_n is a connected graph with $n \geq 3$ vertices where:

$$v_1 - v_2 - \cdots - v_n - v_1$$

forms a closed loop.



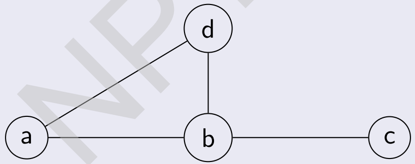
This is the cycle C_4.

Degrees of Vertices

The **degree** of a vertex v, denoted $\deg(v)$, is the number of edges incident to v. One can check that

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Example: In the graph below, $\deg(a) = 2$, $\deg(b) = 3$, $\deg(c) = 1$.



Adjacency Matrix

For a graph $G = (V, E)$ with $V = \{v_1, \ldots, v_n\}$, its **adjacency matrix** $A = [a_{ij}]$ is defined by:

$$a_{ij} = \begin{cases} 1, & \text{if } \{v_i, v_j\} \in E, \\ 0, & \text{otherwise.} \end{cases}$$

Example:
For a path $1 - 2 - 3$,

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Random Graph

Random graph model is a probability distribution on a set of graphs.

Erdős–Rényi Random Graph Model

The Erdős–Rényi random graph is a probability distribution on graphs with n vertices.
Two classical versions:

- $G(n, p)$: each edge is included independently with probability p, where p depends on n.
- $G(n, M)$: a graph chosen uniformly from all graphs with n vertices and exactly M edges.

Typical notation:

$$G \sim G(n, p).$$

The Model $G(n,p)$

In the $G(n,p)$ model:

- Vertex set is fixed: $V = \{1, 2, \ldots, n\}$.
- Each of the $\binom{n}{2}$ possible edges is independently selected.
- Probability that a specific graph with m edges occurs is

$$p^m (1-p)^{\binom{n}{2} - m}.$$

Expected number of edges:

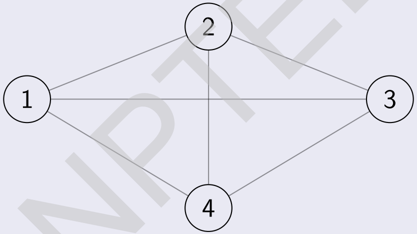
$$\mathbb{E}[|E|] = p\binom{n}{2}.$$

Proof : Here the random variable $|E|$, the number of edges in the random graph, follows the binomial $\left(\binom{n}{2}, p\right)$ distribution. Therefore,

$$\mathbb{E}(|E|) = \binom{n}{2} p.$$

Small Example of $G(n, p)$

Example: $G(4, p)$.
There are 4 vertices and 6 possible edges. Each appears with probability p.



Each edge appears independently with probability p.

Typical Structure: Sparse vs Dense

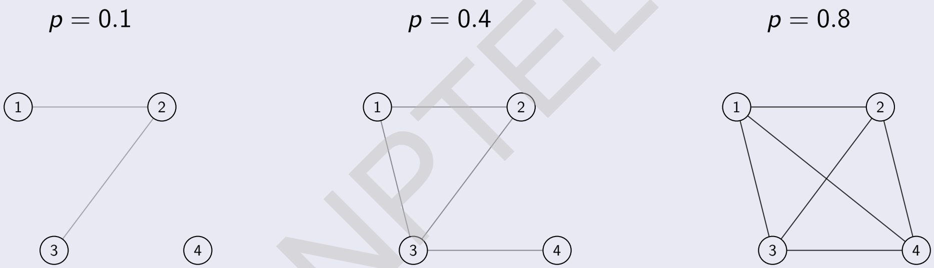
Sparse regime: $p = c/n$

- Average degree $\approx c$.
- If $c < 1$: only small components.
- If $c > 1$: giant component forms.

Dense regime: $p = c$ (constant)

- Graph behaves like a dense random graph.

Example Realizations at Different p and $n = 4$



Connectivity in $G(n, p)$

A central result:

$$G(n,p) \text{ is connected with high probability as } n \to \infty \iff p \geq \frac{\ln n + c(n)}{n},$$

where $c(n) \to \infty$.
At $p = \frac{\ln n}{n}$, a sharp transition occurs:

- Below: many isolated vertices.
- Above: graph becomes connected.

Cliques in $G(n, p)$

Let X_k be the number of k-cliques in $G(n, p)$.

$$\mathbb{E}[X_k] = \binom{n}{k} p^{\binom{k}{2}}.$$

When $p = 1/2$, the largest clique size is about

$$(2 + o(1)) \log_2 n.$$

Adjacency Matrix of $G(n, p)$

The adjacency matrix A of $G(n, p)$ satisfies:

$$A_{ij} = \begin{cases} 1 & \text{with probability } p, \\ 0 & \text{with probability } 1 - p, \end{cases}$$

independently for all $i < j$.

Expected number of ones:

$$\mathbb{E}\left[\text{sum of entries of } A\right] = 2 \cdot p \binom{n}{2}.$$

Limiting Behavior

Let $G = (V, E)$ be a connected graph. For any two vertices $u, v \in V$, let $d(u, v)$ denote the length of the shortest path between u and v. The **diameter** of G, denoted by $\text{diam}(G)$, is defined as

$$\text{diam}(G) = \max_{u,v \in V} d(u, v).$$

Result: If $p > \frac{(1+\varepsilon)\ln n}{n}$, diameter is 2 or 3 with high pobability as $n \to \infty$.

Relation Between Random Graphs and Bernoulli Distribution

In the Erdős–Rényi random graph model $G(n, p)$:

- There are n labelled vertices.
- Every possible edge appears independently with probability p.

This means:

$$\mathbb{P}[(i, j) \text{ is an edge}] = p.$$

Bernoulli View of Edges

For each pair of vertices (i, j) define a random variable:

$$X_{ij} = \begin{cases} 1 & \text{if edge } (i, j) \text{ exists,} \\ 0 & \text{otherwise.} \end{cases}$$

Then:

$$X_{ij} \sim \text{Bernoulli}(p), \quad \text{independent over all } i < j.$$

$$G(n, p) \iff \{X_{ij}\}_{i<j}.$$

References

1. Bollobás, B. Random Graphs, 2nd ed., Cambridge University Press, Cambridge, 2001.
2. Janson, S., Łuczak, T., and Ruciński, A. Random Graphs, Wiley-Interscience, New York, 2000.
3. Frieze, A., and Karoński, M. Introduction to Random Graphs, Cambridge University Press, Cambridge, 2016.
4. Spencer, J. The Strange Logic of Random Graphs, Springer-Verlag, New York, 2001.


Thank you

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Module 11: Random Graph

Lecture 02: Properties of Random Graphs

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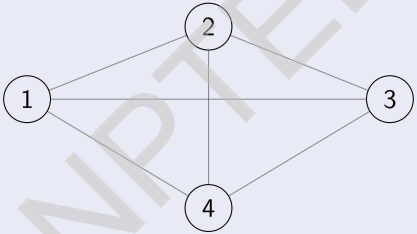

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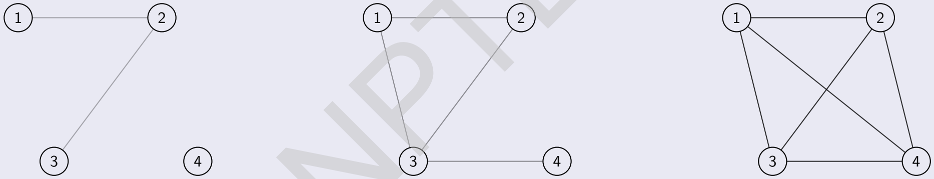
$p = 0.1$ $p = 0.4$ $p = 0.8$



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$$G(n, p) \iff \{X_{ij}\}_{i<j}.$$

Edge Probability and Sparsity

- In $G(n, p)$, each pair of vertices forms an edge independently with probability p, where p depends on n.
- Expected number of edges: $\mathbb{E}[|E|] = \binom{n}{2} p$.
- Sparse regime: $p = \frac{c}{n}$; Dense regime: p constant.
- Threshold phenomena depend strongly on how p scales with n.

Degree Distribution

- Degree of each vertex: Binomial$(n-1, p)$.
- Expected degree: $\mathbb{E}[\deg(v)] = (n-1)p$.
- For $p = \frac{c}{n}$, degrees converge to Poisson(c) as $n \to \infty$.
- High concentration around the mean when $np \gg \log n$.

Connectivity Threshold

- A sharp threshold for connectivity occurs at $p = \frac{\log n}{n}$.
- If $p = \frac{\log n - \omega(1)}{n}$, graph is disconnected with high probability as $n \to \infty$, where an integer-valued function $f(n)$ is said to be an $\omega(1)$ function if $f(n) \to \infty$ as $n \to \infty$.
- If $p = \frac{\log n + \omega(1)}{n}$, graph is connected with high probability as $n \to \infty$.
- Main obstruction to connectivity around threshold is isolated vertices.

Triangles

- Expected number of triangles: $\mathbb{E}[T] = \binom{n}{3} p^3$.

  Proof : Let $G(n, p)$ be an Erdős–Rényi random graph and let T denote the number of triangles in the graph. There are $\binom{n}{3}$ possible triples of vertices. For each triple T_i, define the indicator variable

$$I_i = \begin{cases} 1, & \text{if the three vertices of } T_i \text{ form a triangle,} \\ 0, & \text{otherwise.} \end{cases}$$

  A triple forms a triangle if all three of its edges are present, each occurring independently with probability p; thus

$$\mathbb{E}[I_i] = p^3.$$

  Since

$$T = \sum_{i=1}^{\binom{n}{3}} I_i,$$

  by linearity of expectation,

$$\mathbb{E}[T] = \sum_{i=1}^{\binom{n}{3}} \mathbb{E}[I_i] = \binom{n}{3} p^3 = \frac{n(n-1)(n-2)}{6} p^3.$$

Triangles (continued)

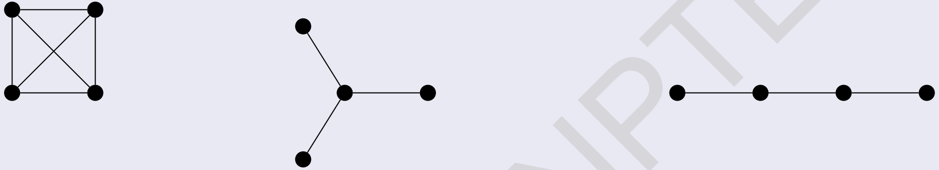
- Random graphs have no local structure; triangles appear randomly.
- Contrasts with social-network graphs which have high clustering.

Diameter and Distances

- Diameter $\approx \frac{\log n}{\log(np)}$, when $np \gg \log n$.
- If p is constant, diameter is almost surely 2.
- In sparse regime $p = \frac{c}{n}$, typical distances scale as $\log n$.

Example

Suppose for an Erdős–Rényi random graph model $G(n, p)$, one has the data for the following three graphs. Calculate the maximum likelihood estimate of p.



Solution

Observe that,

$$\mathbf{P}(G(n,p) = K_4) = p^6$$
$$\mathbf{P}(G(n,p) = K_{1,3}) = p^3(1-p)^3$$
$$\mathbf{P}(G(n,p) = P_4) = p^3(1-p)^3$$

The likelihood function is

$$L = p^6 \times p^3(1-p)^3 \times p^3(1-p)^3$$
$$= p^{12}(1-p)^6$$

Solution (continued)

Taking log both sides we get,

$$\log(L) = 12\,\log(p) + 6\,\log(1-p)$$

Differentiating both sides with respect to p,

$$\frac{\partial \log(L)}{\partial p} = \frac{12}{p} - \frac{6}{1-p}$$

Solution (continued)

Now equating $\frac{\partial \log(L)}{\partial p} = 0$ gives

$$\frac{12}{p} = \frac{6}{1-p}$$
$$\text{or, } 2(1-p) = p$$
$$\text{or, } p = \frac{2}{3}$$

One can show that by second derivative test that the value of $\frac{\partial^2 \log(L)}{\partial p^2}$ ar $p = \frac{2}{3}$ is negative. Therefore, the maximum likelihood estimate of p is $\hat{p} = \frac{2}{3}$

Maximum Likelihood Estimate of Random Graph

Consider an undirected graph $G = (V, E)$ on n vertices that is assumed to be a single realization of the Erdős–Rényi random graph model $G(n, p)$, where each possible edge appears independently with probability $p \in [0, 1]$.
Number of possible edges:

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

Let $m = |E|$ be the observed number of edges in G.
The probability of observing a specific graph with exactly m edges is

$$P(G \mid p) = p^m (1-p)^{\binom{n}{2}-m}$$

because there are m edges that must be present and $\binom{n}{2} - m$ edges that must be absent, and all edge indicators are independent.

Maximum Likelihood Estimate of Random Graph (continued)

The log-likelihood function is

$$\begin{aligned}\ell(p) &= \log \mathrm{P}(G \mid p) \\ &= \log \left[p^m (1-p)^{\binom{n}{2}-m}\right] \\ &= m \log p + \left(\binom{n}{2} - m\right) \log(1-p).\end{aligned}$$

Differentiate with respect to p:

$$\begin{aligned}\frac{d\ell}{dp} &= \frac{\partial}{\partial p}\left[m \log p + \left(\binom{n}{2} - m\right) \log(1-p)\right] \\ &= \frac{m}{p} - \frac{\binom{n}{2} - m}{1-p}.\end{aligned}$$

Maximum Likelihood Estimate of Random Graph (continued)

Set the score equal to zero:

$$\frac{m}{p} = \frac{\binom{n}{2} - m}{1 - p}$$

Cross-multiply:

$$m(1 - p) = \left(\binom{n}{2} - m\right) p$$

Expand:

$$m - mp = \binom{n}{2} p - mp$$

Maximum Likelihood Estimate of Random Graph (continued)

The $-mp$ terms cancel:

$$m = \binom{n}{2} p$$

Solve for p:

$$\hat{p}_{\text{MLE}} = \frac{m}{\binom{n}{2}} = \frac{m}{\frac{n(n-1)}{2}} = \frac{2m}{n(n-1)}$$

$$\frac{d^2\ell}{dp^2} = -\frac{m}{p^2} - \frac{\binom{n}{2} - m}{(1-p)^2} < 0 \quad \text{for } 0 < p < 1,\ m \in \left\{0, 1, \ldots, \binom{n}{2}\right\}$$

The Hessian is strictly negative, so $\hat{p}_{\text{MLE}}$ is indeed a maximum.

Example

Consider an observed simple undirected graph on $n = 10$ vertices, containing $m = 12$ edges. Assume the model $G(n, p)$, in which each of the $\binom{n}{2}$ possible edges appears independently with probability p. Find the maximum likelihood estimator (MLE) of p.

Solution

The number of possible edges is

$$N = \binom{10}{2} = \frac{10 \cdot 9}{2} = 45.$$

The likelihood for observing exactly m edges is

$$L(p) = p^m (1-p)^{N-m}.$$

Taking the log-likelihood,

$$\ell(p) = m \log p + (N-m)\log(1-p).$$

Differentiate and set to zero:

$$\ell'(p) = \frac{m}{p} - \frac{N-m}{1-p} = 0 \quad \Rightarrow \quad \hat{p} = \frac{m}{N}.$$

Thus, numerically,

$$\hat{p} = \frac{12}{45} = \frac{4}{15} \approx 0.26667.$$

Solution (continued)

To check that this is a maximum, compute the second derivative:

$$\ell''(p) = -\frac{m}{p^2} - \frac{N-m}{(1-p)^2}.$$

At $\hat{p} = \frac{4}{15}$,

$$\ell''(\hat{p}) \approx -230.11 < 0,$$

confirming the maximum.

Final Answer:

$$\boxed{\hat{p} = \frac{4}{15} \approx 0.26667}.$$

References

1. Bollobás, B. Random Graphs, 2nd ed., Cambridge University Press, Cambridge, 2001.
2. Janson, S., Łuczak, T., and Ruciński, A. Random Graphs, Wiley-Interscience, New York, 2000.
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Thank you

SWAYAM NPTEL COURSE ON

STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

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INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 11: Random Graph

Lecture 03: Law of Large Numbers for Random Graph 1

Markov's Inequality

Let $X \geq 0$ be a non–negative random variable (discrete or continuous), and let $k > 0$. Then:

$$P(X \geq k) \leq \frac{\mathbb{E}[X]}{k}.$$

Equivalently (plugging in $k\mathbb{E}[X]$ for k above):

$$P(X \geq k\,\mathbb{E}[X]) \leq \frac{1}{k}.$$

Proof

Below is the proof when X is continuous. The proof for discrete random variables is similar (just replace the integrals by summations).

$$\mathbb{E}[X] = \int_0^\infty x f_X(x)\, dx \qquad \text{(because } X \geq 0\text{)}$$

Split the integral at some $0 \leq k \leq \infty$:

$$\mathbb{E}[X] = \int_0^k x f_X(x)\, dx + \int_k^\infty x f_X(x)\, dx$$

Since $x \geq 0$ and $f_X(x) \geq 0$, the first integral is nonnegative:

$$\mathbb{E}[X] \geq \int_k^\infty x f_X(x)\, dx$$

Proof (continued)

Now, on the region of integration $x \geq k$, so we may replace x by k:

$$\int_k^\infty x f_X(x)\,dx \;\geq\; \int_k^\infty k f_X(x)\,dx = k\int_k^\infty f_X(x)\,dx$$

But

$$\int_k^\infty f_X(x)\,dx = \mathrm{P}(X \geq k),$$

so we obtain:

$$\mathbb{E}[X] \;\geq\; k\,\mathrm{P}(X \geq k).$$

Dividing both sides by $k > 0$ gives the Markov bound:

$$\mathrm{P}(X \geq k) \;\leq\; \frac{\mathbb{E}[X]}{k}.$$

Chebyshev's Inequality

Let X be a random variable with finite mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathrm{Var}(X)$. Then for every $\varepsilon > 0$,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Chebyshev's Inequality: Proof

Consider the nonnegative random variable

$$Y = (X - \mu)^2.$$

By Markov's inequality, for any $a > 0$,

$$P(Y \geq a) \leq \frac{\mathbb{E}[Y]}{a}.$$

Choose $a = \varepsilon^2$. Then

$$\mathrm{P}\left((X - \mu)^2 \geq \varepsilon^2\right) \leq \frac{\mathbb{E}\left[(X - \mu)^2\right]}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}.$$

Notice that

$$\mathrm{P}\left(|X - \mu| \geq \varepsilon\right) = \mathrm{P}\left((X - \mu)^2 \geq \varepsilon^2\right),$$

so substituting back gives

$$\mathrm{P}\left(|X - \mu| \geq \varepsilon\right) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Weak Law of Large Numbers : Statement

Let $\{X_i\}_{i\geq 1}$ be independent random variables with common expectation $\mathbb{E}[X_i] = \mu$ and finite variances $\mathrm{Var}(X_i) = \sigma_i^2$. Suppose

$$\frac{1}{n^2}\sum_{i=1}^{n} \sigma_i^2 \longrightarrow 0 \quad \text{as } n \to \infty.$$

Define the sample mean

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Then $\overline{X}_n \xrightarrow{P} \mu$, i.e. for every $\varepsilon > 0$,

$$\mathrm{P}\big(|\overline{X}_n - \mu| > \varepsilon\big) \to 0 \quad \text{as } n \to \infty.$$

Weak Law of Large Numbers : Proof

First compute mean and variance of the sample mean. Linearity of expectation gives

$$\begin{aligned}
&\mathbb{E}[\overline{X}_n] \\
&= \mathbb{E}\left[\frac{X_1 + \cdots + X_n}{n}\right] \\
&= \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] \\
&= \frac{1}{n}\sum_{i=1}^{n} \mu \\
&= \mu.
\end{aligned}$$

Weak Law of Large Numbers : Proof (continued)

Since the X_i are independent,

$$
\begin{aligned}
&\mathrm{Var}(\overline{X}_n) \\
&= \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} \sigma_i^2.
\end{aligned}
$$

Weak Law of Large Numbers : Proof (continued)

By the hypothesis this variance tends to 0 as $n \to \infty$. Now apply Chebyshev's inequality: for any $\varepsilon > 0$,

$$\mathrm{P}\big(|\overline{X}_n - \mu| > \varepsilon\big) \le \frac{\mathrm{Var}(\overline{X}_n)}{\varepsilon^2} = \frac{1}{\varepsilon^2} \cdot \frac{1}{n^2} \sum_{i=1}^{n} \sigma_i^2.$$

The right-hand side tends to 0 by assumption, therefore $P(|\overline{X}_n - \mu| > \varepsilon) \to 0$. This is precisely $\overline{X}_n \xrightarrow{P} \mu$.
For i.i.d. variables $\mathrm{Var}(\overline{X}_n) = \sigma^2/n \to 0$ as $n \to \infty$. Apply Chebyshev's inequality as in the theorem to conclude convergence in probability.

A Weak Law of Large Numbers of Random Graph

Given n vertices labelled by $\{1, 2, \ldots, n\}$, independently for each unordered pair of vertices $\{i, j\}$ with $1 \leq i \neq j \leq n$. Draw an edge between i and j with a fixed probability $p \in (0, 1)$. A set of three vertices $\{i, j, k\}$ is said to form a triangle if there is an edge between every pair of vertices. Let Δ_n denote the number of triangles in the random graph. Then

$$\mathrm{P}\left(\left| \frac{\Delta_n}{E(\Delta_n)} - 1 \right| > \varepsilon \right) \to 0 \quad \text{as } n \to \infty$$

References

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Module 11: Random Graph

Lecture 04: Law of Large Numbers for Random Graph 2

Chebyshev's Inequality

Let X be a random variable with finite mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathrm{Var}(X)$. Then for every $\varepsilon > 0$,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Indicator Function

Let $A \subseteq X$. The indicator function of A is defined as

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \notin A. \end{cases}$$

A Weak Law of Large Numbers of Random Graph

Given n vertices labelled by $\{1, 2, \ldots, n\}$, independently for each unordered pair of vertices $\{i, j\}$ with $1 \leq i \neq j \leq n$. Draw an edge between i and j with a fixed probability $p \in (0, 1)$. A set of three vertices $\{i, j, k\}$ is said to form a triangle if there is an edge between every pair of vertices. Let Δ denote the number of triangles in the random graph. Then

$$P\left(\left|\frac{\Delta}{E(\Delta)} - 1\right| > \varepsilon\right) \to 0 \quad \text{as } n \to \infty$$

Proof

1. Indicator Representation

For each unordered triple $1 \le i < j < k \le n$, define the indicator

$$\Delta_{ijk} = \mathbf{1}\{\{i,j\},\{j,k\},\{k,i\} \text{ are edges}\}.$$

Then

$$\Delta = \sum_{i<j<k} \Delta_{ijk}, \qquad \#\{\text{triangles}\} = \binom{n}{3}.$$

2. Expectation

Since three independent edges must be present,

$$\mathbb{E}[\Delta_{ijk}] = p^3.$$

Thus

$$\mathbb{E}[\Delta] = \binom{n}{3} p^3.$$

Proof (continued)

3. Variance Decomposition

$$\mathrm{Var}(\Delta) = \sum_t \mathrm{Var}(\Delta_t) + \sum_{t \neq s} \mathrm{Cov}(\Delta_t, \Delta_s).$$

where t and s are tuples (triplets).

3.1 Variance of a Single Indicator

$$\mathrm{Var}(\Delta_{ijk}) = p^3 - p^6 = p^3(1 - p^3).$$

So

$$\sum_t \mathrm{Var}(\Delta_t) = \binom{n}{3} p^3(1 - p^3).$$

Proof

3.2 Covariances

Two triangle indicators are independent unless the triangles share exactly one edge.

Case: Two triangles share one edge.

Example: $\{i, j, k\}$ and $\{i, j, \ell\}$ share edge $\{i, j\}$.

They require exactly 5 distinct edges, so:

$$\mathbb{E}[\Delta_t \Delta_s] = p^5, \qquad \mathbb{E}[\Delta_t]\mathbb{E}[\Delta_s] = p^6.$$

Therefore

$$\mathrm{Cov}(\Delta_t, \Delta_s) = p^5 - p^6 = p^5(1 - p).$$

Proof (continued)

Counting such pairs.
Choose the shared edge: $\binom{n}{2}$ ways.
Pick two distinct third vertices: $\binom{n-2}{2}$ ways.
Thus number of unordered pairs of triangles sharing one edge:

$$\binom{n}{2}\binom{n-2}{2}.$$

4. Final Variance Formula
Combining all contributions:

$$\boxed{\mathrm{Var}(\Delta) = \binom{n}{3}p^3(1-p^3) + \binom{n}{2}\binom{n-2}{2}p^5(1-p)}$$

Proof (continued)

By Chebyshev's inequality

$$\mathrm{P}(|\Delta - E[\Delta]| > \varepsilon E[\Delta]) \leq \frac{\mathrm{Var}(\Delta)}{\varepsilon^2 E[\Delta]^2}$$

$$\text{or, } \mathrm{P}\left(\left|\frac{\Delta}{E[\Delta]} - 1\right| > \varepsilon\right) \leq \frac{\mathrm{Var}(\Delta)}{\varepsilon^2 E[\Delta]^2}$$

Now,

$$\frac{\mathrm{Var}(\Delta)}{E[\Delta]^2} = \frac{\binom{n}{3} p^3 (1 - p^3) + \binom{n}{2}\binom{n-2}{2} p^5 (1-p)}{\binom{n}{3}^2 p^6}$$

Proof (continued)

Observe that the numerator contains n^4 the highest degree term in n whereas the denominator contains n^6 as the highest degree term in n. Therefore when $n \to \infty$,

$$\frac{\mathrm{Var}(\Delta)}{E[\Delta]^2} \to 0$$

and the proof is complete.

References

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Module 11: Introduction

Lecture 05: Law of Large Numbers for Random Graph 3

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$$P\left(\left|\frac{\Delta}{E(\Delta)} - 1\right| > \varepsilon\right) \to 0 \quad \text{as } n \to \infty$$

Proof

1. Indicator Representation

For each unordered triple $1 \le i < j < k \le n$, define the indicator

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